INCH-POUND

MIL-DTL-3933/14D
7 November 2002
SUPERSEDING
MIL-A-3933/14C
24 April 1985

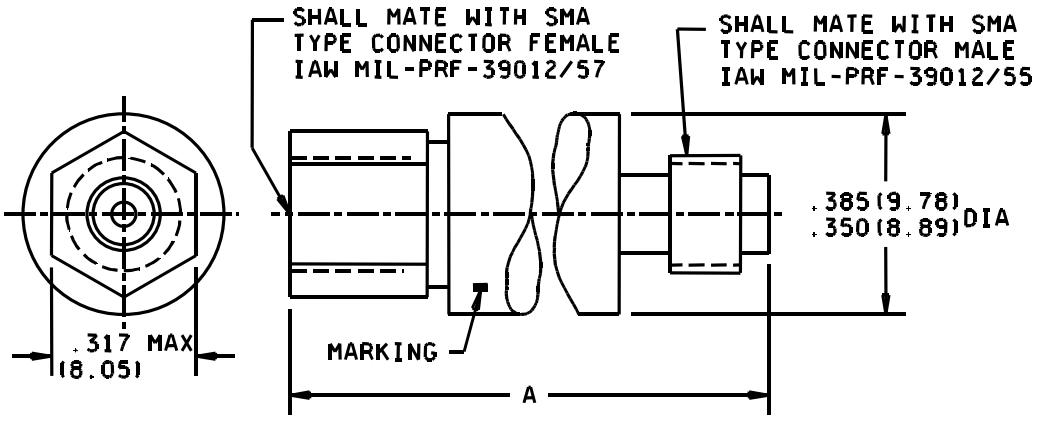
DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SMA CONNECTORS),

FREQUENCY RANGE: DC TO 12.4 GHz, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING DATA:

Attenuators for use in equipment in which precision and stability are secondary considerations, except for temperature sensitivity.

Operating frequency range: dc to 12.4 GHz.

Test frequency range for SWR and attenuation measurements, measured or recorded, when using Automatic Network Analyzer (ANA) or equivalent test equipment:

100 MHz (or lower) to 12.4 GHz.

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' spec is imposed: 6.0 GHz

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	up to 10 dB, inclusive (dB)	over 10 dB (dB/dB)
temperature change or thermal shock	0.5	0.05
vibration or shock	0.5	0.05
moisture resistance or salt spray	0.5	0.05
peak power	0.5	0.05

Frequency sensitivity, maximum: 0.4 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.04 dB.

Part or Identifying Number (PIN): M3933/14- (and dash number from table I).

Supersession data: See table II.

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TABLE I. Electrical characteristics.

Dash number		Attenuation dB			Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation				dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
				DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)					
01	N S	3		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
02	N S	6		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
03	N S	10		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
04	N S	20		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
05	N S	15		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
06	N S	1		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
07	N S	2		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
08	N S	4		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
09	N S	5		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
10	N S	7		±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
11	N S	8		±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
12	N S	9		±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
13	N S	30		±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB			Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation				dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
				DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)					
14	N S	40		±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
15	N S	60		±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
16	N S	50		±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
17	N S	28		±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.7	1.850 (46.99)
18	N S	16		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
19	N S	14		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
20	N S	13		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
21	N S	12		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
22	N S	11		±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
23	N S	1.5		±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
24	N S	31		±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration 5 microseconds.

3/ Power input is derated linearly from +25°C to 0.5 watts at +125°C.

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TABLE II. Supersession data.

PIN M3933/14-	Supersedes M3933/15-
01	01
02	02
03	03
04	04
13	05
14	06
13	07
14	08

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, MC
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-1229-02)

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